

Suggested FYP topics for 2026-27

by

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Project code :	YJZ1
Project title :	Reinforcement learning for next-generation wireless communication systems
No. of students :	2
Description of project :	Reinforcement Learning (RL) is an advanced machine learning paradigm that enables agents to learn optimal strategies by interacting with an environment. In the field of wireless communication, RL has emerged as a powerful tool to address complex challenges such as dynamic resource allocation, spectrum management, and etc.. Unlike traditional rule-based methods, RL allows systems to adapt intelligently to changing network conditions, making it a key enabler for the next generation of wireless communication systems. In this project, students are expected to design and implement RL algorithms to optimize a specific aspect of a wireless communication system. They will simulate a wireless environment, apply RL techniques to improve system performance, and evaluate the effectiveness of their approach. Key aspects to investigate: • What to optimize: Students can explore different optimization objectives, such as improving spectrum efficiency, reducing energy consumption, or maximizing coverage. They will evaluate the impact of these objectives on overall system performance. • How to learn: Students can study and compare different RL algorithms, e.g., Q-learning, Deep Q-Networks (DQN), and Policy Gradient methods, to determine their effectiveness in solving specific wireless communication challenges. They will also investigate the effect of hyperparameters such as learning rate, reward function design, and exploration-exploitation trade-offs. • Real-world application: Students can simulate real-world scenarios to test the robustness and adaptability of their RL solutions. Students taking this project are expected to have strong programming skills and a basic understanding of machine learning concepts. By the end of the project, they will gain hands-on experience in applying RL to a critical real-world domain, preparing them for future research or industry opportunities in wireless communication systems.
Industrial collaboration :	No.

Project code :	YJZ2
Project title :	Radio Environment Mapping with Machine Learning
No. of students :	2
Description of project :	In this project, students will create a radio map of a wireless signal coverage area and use machine learning to predict signal strength at unmeasured locations. In other words, students will simulate (or measure) the signal strength at some points in a small indoor/outdoor area and then build a model that “fills in” the coverage map. This teaches how wireless signal propagation works and how AI can help estimate coverage. Through the project, students will learn • Wireless Concepts: Understand how signal strength decreases with distance (path loss) and how real environments cause variability. • Data Handling: Work with a small dataset and practice cleaning/structuring data for ML. • Machine Learning: Learn how to train a regression/prediction model. Start with a simple interpolator or regression, understanding how it picks up patterns, and possibly extend to more complex models. • Evaluation and Visualization: Use quantitative metrics and visual tools (heatmaps, contour plots) to evaluate your model’s accuracy. Presenting results visually is a key skill. Students are expected to be strong at python programming. A basic understanding of communications systems is preferable.
Industrial collaboration :	No